

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
REPORT TEMPLAT

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Code & Course	:	DFP40443 FULL STACK WEB DEVELOPMENT	Class	:	DIT4B
Lecturer	:	MR. MOHD AZLAN BIN ABDUL AZIZ	Date	:	02 APRIL 2026
Assesment	:	MINI PROJECT			

Put your report inside this.

1.	Database Design & Structure	Constructs and optimizes well-structured relational database with clear relationships and normalization	Constructs structured tables with correct relationships (PK/FK)	Constructs basic tables but relationships are unclear or weak	Constructs incomplete tables with poor field selection; no relationships	2	
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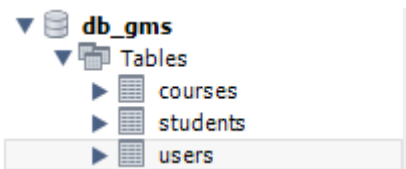


Table: users

Columns:

id int(11) AI PK
username varchar(50)
password varchar(255)
role enum('Admin','User')

Table: students

Columns:

id int(11) AI PK
reg_no varchar(20)
name varchar(100)
email varchar(100)
course_id int(11)

Table: courses

Columns:

course_id int(11) AI PK
course_name varchar(100)

2.

Database Connection (PHP-MySQL)	Configures robust and reusable database connection with proper error handling	Implements stable connection using correct configuration	Establishes connection with frequent errors	Fails to establish connection consistently	2	
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MiniProject1 > config > db.php

You, yesterday | 1 author (You)

```

1  <?php
2  $host = "127.0.0.1:3307";
3  $user = "root";
4  $pass = "";
5  $dbname = "db_gms";
6
7  $conn = new mysqli($host, $user, $pass, $dbname);
8
9  if ($conn->connect_error) {
10     die("Connection failed: " . $conn->connect_error);
11 }
12 ?>
```

3.

Authentication System (Register, Login, Logout, Session)	Integrates secure authentication system with proper validation, session control, and logout handling	Implements functional authentication with session management	Implements basic login/register with limited session control	Implements incomplete authentication; session handling fails	2	
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<<Login.php>>

```

<?php
session_start();
include 'config/db.php';

if (isset($_POST['login'])) {
    $username = $_POST['username'];
    $password = $_POST['password'];

    $stmt = $conn->prepare("SELECT role FROM users WHERE username=? AND password=?");
    $stmt->bind_param("ss", $username, $password);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows == 1) {
        $user = $result->fetch_assoc();
        $_SESSION['username'] = $username;
        $_SESSION['role'] = $user['role'];
        header("Location: dashboard.php");
    } else {
        $error = "Invalid Username or Password";
    }
}
?>
```

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login - SMS</title>
    <link
href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.cs
s" rel="stylesheet">
</head>
<body class="bg-light">
    <div class="container mt-5">
        <div class="row justify-content-center">
            <div class="col-md-4">
                <div class="card shadow">
                    <div class="card-header bg-primary text-white text-
center"><h4>Login</h4></div>
                    <div class="card-body">
                        <?php if(isset($error)) echo "<div class='alert
alert-danger'>$error</div>"; ?>
                        <form method="POST">
                            <div class="mb-3">
                                <label>Username</label>
                                <input type="text" name="username"
class="form-control" required>
                            </div>
                            <div class="mb-3">
                                <label>Password</label>
                                <input type="password" name="password"
class="form-control" required>
                            </div>
                            <button type="submit" name="login" class="btn
btn-primary w-100">Login</button>
                        </form>
                    </div>
                </div>
            </div>
        </div>
    </div>
</body>
</html>

```

<<logout.php>>

```

<?php
session_start();
session_destroy();
header("Location: login.php");
?>

```

4.

Role-Based Access (Admin/User)	Integrates complete role-based control with secure restriction of features	Implements role-based access correctly for most functions	Differentiates roles but with incorrect access control	Fails to differentiate roles	2	
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<<manage_students.php>>

ROLE-BASED ACCESS (ADMIN)

```
<?php if($_SESSION['role'] == 'Admin'): ?>
    <a href="add_student.php" class="btn btn-primary">+ Add
Student</a>
<?php endif; ?>
```

```
<?php if($_SESSION['role'] == 'Admin'): ?>
    <a href="edit_student.php?id=<?php echo
$row['id']; ?>" class="btn btn-warning btn-sm">Edit</a>
    <a href="delete_student.php?id=<?php echo
$row['id']; ?>" class="btn btn-danger btn-sm" onclick="return confirm('Are
you sure?')">Delete</a>
<?php endif; ?>
```

*User have limited access, so user can only view dashboard & student name list on 'Manage Student'. Only Admin can Add, Edit, Change & Delete students.

5.

CRUD Operations (Create, Read, Update, Delete)	Constructs and integrates complete CRUD system with efficient logic and clean execution	Implements full CRUD operations correctly	Implements basic CRUD with errors or missing features	Implements limited or non-functional CRUD operations	2	
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<<add_student.php>> (ADD)

```
<?php
session_start();
include 'config/db.php';
if($_SESSION['role'] != 'Admin') header("Location: manage_students.php");

if (isset($_POST['add'])) {
    $reg = $_POST['reg_no'];
    $name = $_POST['name'];
    $email = $_POST['email'];
    $course = $_POST['course_id'];

    $stmt = $conn->prepare("INSERT INTO students (reg_no, name, email,
course_id) VALUES (?, ?, ?, ?)");
    $stmt->bind_param("sssi", $reg, $name, $email, $course);

    if ($stmt->execute()) {
        header("Location: manage_students.php?success=1");
    }
}
$courses = $conn->query("SELECT * FROM courses");
?>

<!DOCTYPE html>
```

```

<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Add Student</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.cs
s" rel="stylesheet">
</head>
<body class="p-5">
    <div class="container col-md-6 border p-4 bg-white shadow">
        <h3>Add New Student</h3>
        <form method="POST">
            <div class="mb-3"><label>Registration No</label><input
type="text" name="reg_no" class="form-control" required></div>
            <div class="mb-3"><label>Full Name</label><input type="text"
name="name" class="form-control" required></div>
            <div class="mb-3"><label>Email</label><input type="email"
name="email" class="form-control" required></div>
            <div class="mb-3">
                <label>Course</label>
                <select name="course_id" class="form-select">
                    <?php while($c = $courses->fetch_assoc()): ?>
                        <option value="<?php echo $c['course_id']; ?>"><?php
echo $c['course_name']; ?></option>
                    <?php endwhile; ?>
                </select>
            </div>
            <button type="submit" name="add" class="btn btn-success">Save
Student</button>
            <a href="manage_students.php" class="btn btn-link">Cancel</a>
        </form>
    </div>
</body>
</html>

```

<<manage_students.php>> (READ)

```

<?php
session_start();
include 'config/db.php';

$search = $_GET['search'] ?? '';
$query = "SELECT students.*, courses.course_name
FROM students
JOIN courses ON students.course_id = courses.course_id
WHERE students.name LIKE ? OR students.reg_no LIKE ?
ORDER BY students.id DESC";

$stmt = $conn->prepare($query);
$search_term = "%$search%";

```

```

$stmt->bind_param("ss", $search_term, $search_term);
$stmt->execute();
$result = $stmt->get_result();
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Manage Students</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.cs
s" rel="stylesheet">
</head>
<body class="bg-light">
    <div class="container mt-4">
        <div class="d-flex justify-content-between align-items-center mb-3">
            <h3>Student Records</h3>
            <form class="d-flex w-50">
                <input type="text" name="search" class="form-control me-2"
placeholder="Search by name or Reg No..." value="<?php echo $search; ?>">
                <button type="submit" class="btn btn-outline-
primary">Search</button>
            </form>
            <?php if($_SESSION['role'] == 'Admin'): ?>
                <a href="add_student.php" class="btn btn-primary">+ Add
Student</a>
            <?php endif; ?>
        </div>

        <table class="table table-bordered table-striped bg-white shadow-
sm">
            <thead class="table-dark">
                <tr>
                    <th>Reg No</th>
                    <th>Name</th>
                    <th>Email</th>
                    <th>Course</th>
                    <th>Actions</th>
                </tr>
            </thead>
            <tbody>
                <?php while($row = $result->fetch_assoc()): ?>
                    <tr>
                        <td><?php echo $row['reg_no']; ?></td>
                        <td><?php echo $row['name']; ?></td>
                        <td><?php echo $row['email']; ?></td>
                        <td><?php echo $row['course_name']; ?></td>

```

```

        <td>
            <?php if($_SESSION['role'] == 'Admin'): ?>
                <a href="edit_student.php?id=?php echo
$row['id']; ?>" class="btn btn-warning btn-sm">Edit</a>
                <a href="delete_student.php?id=?php echo
$row['id']; ?>" class="btn btn-danger btn-sm" onclick="return confirm('Are
you sure?')">Delete</a>
            <?php endif; ?>
        </td>
    </tr>
    <?php endwhile; ?>
</tbody>
</table>
<a href="dashboard.php" class="btn btn-secondary">Back to
Dashboard</a>
</div>
</body>
</html>

```

<<edit_student.php>> (UPDATE)

```

<?php
session_start();
include 'config/db.php';

// Security: Only Admin can access this page
if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'Admin') {
    header("Location: manage_students.php");
    exit();
}

// Fetch existing student data
if (isset($_GET['id'])) {
    $id = $_GET['id'];
    $stmt = $conn->prepare("SELECT * FROM students WHERE id = ?");
    $stmt->bind_param("i", $id);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows === 1) {
        $student = $result->fetch_assoc();
    } else {
        header("Location: manage_students.php?error=StudentNotFound");
        exit();
    }
} else {
    header("Location: manage_students.php");
    exit();
}

```

```

// Handle form submission for updating
if (isset($_POST['update'])) {
    $reg = $_POST['reg_no'];
    $name = $_POST['name'];
    $email = $_POST['email'];
    $course = $_POST['course_id'];
    $student_id = $_POST['id'];

    $update_stmt = $conn->prepare("UPDATE students SET reg_no=?, name=?,
email=?, course_id=? WHERE id=?");
    $update_stmt->bind_param("sssi", $reg, $name, $email, $course,
$student_id);

    if ($update_stmt->execute()) {
        header("Location: manage_students.php?success=StudentUpdated");
        exit();
    } else {
        $error = "Error updating record: " . $conn->error;
    }
}

// Fetch courses for the dropdown
$courses = $conn->query("SELECT * FROM courses");
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Edit Student</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.cs
s" rel="stylesheet">
</head>
<body class="bg-light p-5">
    <div class="container col-md-6 border p-4 bg-white shadow rounded">
        <h3 class="mb-4">Edit Student Record</h3>

        <?php if(isset($error)) echo "<div class='alert alert-
danger'>$error</div>"; ?>

        <form method="POST">
            <input type="hidden" name="id" value="<?php echo $student['id'];
?>">

            <div class="mb-3">
                <label class="form-label">Registration No</label>

```

```

        <input type="text" name="reg_no" class="form-control"
value="<?php echo htmlspecialchars($student['reg_no']); ?>" required>
    </div>

    <div class="mb-3">
        <label class="form-label">Full Name</label>
        <input type="text" name="name" class="form-control"
value="<?php echo htmlspecialchars($student['name']); ?>" required>
    </div>

    <div class="mb-3">
        <label class="form-label">Email</label>
        <input type="email" name="email" class="form-control"
value="<?php echo htmlspecialchars($student['email']); ?>" required>
    </div>

    <div class="mb-3">
        <label class="form-label">Course</label>
        <select name="course_id" class="form-select" required>
            <?php while($c = $courses->fetch_assoc()): ?>
                <option value="<?php echo $c['course_id']; ?>" <?php
if($student['course_id'] == $c['course_id']) echo 'selected'; ?>>
                    <?php echo htmlspecialchars($c['course_name']);
?>
                </option>
            <?php endwhile; ?>
        </select>
    </div>

    <button type="submit" name="update" class="btn btn-warning text-
dark fw-bold">Update Student</button>
    <a href="manage_students.php" class="btn btn-secondary ms-
2">Cancel</a>
</form>
</div>
</body>
</html>

```

<<delete_student.php>> (DELETE)

```

<?php
session_start();
include 'config/db.php';

// Security: Only Admin can perform deletions
if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'Admin') {
    header("Location: manage_students.php?error=UnauthorizedAction");
    exit();
}

```

```
// Check if an ID was passed in the URL
if (isset($_GET['id'])) {
    $id = $_GET['id'];

    // Use prepared statements to prevent SQL injection
    $stmt = $conn->prepare("DELETE FROM students WHERE id = ?");
    $stmt->bind_param("i", $id);

    if ($stmt->execute()) {
        // Redirect back with a success message
        header("Location: manage_students.php?success=StudentDeleted");
    } else {
        // Redirect back with an error message
        header("Location: manage_students.php?error=DeleteFailed");
    }

    $stmt->close();
} else {
    // If no ID is set, just send them back
    header("Location: manage_students.php");
}

$conn->close();
exit();
?>
```

6.

Advanced SQL Queries (WHERE, JOIN, ORDER BY, GROUP BY)	Integrates complex queries effectively to manipulate and retrieve meaningful data	Implements queries correctly with appropriate clauses	Uses basic queries with limited correctness	Uses incorrect or no query conditions	2	
--	---	---	---	---------------------------------------	---	--

(WHERE, JOIN, ORDER BY) is apply in:

<<manage_students.php>>

```
$query = "SELECT students.*, courses.course_name
FROM students
JOIN courses ON students.course_id = courses.course_id
WHERE students.name LIKE ? OR students.reg_no LIKE ?
ORDER BY students.id DESC";
```

<<edit_student.php>>

```
if (isset($_POST['update'])) {
    $reg = $_POST['reg_no'];
    $name = $_POST['name'];
    $email = $_POST['email'];
    $course = $_POST['course_id'];
    $student_id = $_POST['id'];

    $update_stmt = $conn->prepare("UPDATE students SET reg_no=?, name=?,
email=?, course_id=? WHERE id=?");
    $update_stmt->bind_param("sssii", $reg, $name, $email, $course,
$student_id);
```

7.

Prepared Statements & Security	Integrates secure parameterized queries consistently with strong input validation	Implements prepared statements correctly	Uses prepared statements inconsistently	Does not use prepared statements; vulnerable to SQL injection	2	
--------------------------------	---	--	---	---	---	--

<<login.php>>

User have access as user only, and Admin have access as admin.

```
<?php
session_start();
include 'config/db.php';

if (isset($_POST['login'])) {
    $username = $_POST['username'];
    $password = $_POST['password'];

    $stmt = $conn->prepare("SELECT role FROM users WHERE username=? AND
password=?");
    $stmt->bind_param("ss", $username, $password);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows == 1) {
        $user = $result->fetch_assoc();
        $_SESSION['username'] = $username;
        $_SESSION['role'] = $user['role'];
        header("Location: dashboard.php");
    } else {
        $error = "Invalid Username or Password";
    }
}
?>
```

8.

Search Functionality	Constructs efficient search feature with dynamic filtering and accurate results	Implements functional and accurate search	Basic search implemented but inaccurate	Search function is missing or non-functional	2	
----------------------	---	---	---	--	---	--

<<manage_students.php>>

```
<?php
session_start();
include 'config/db.php';

$search = $_GET['search'] ?? '';
$query = "SELECT students.*, courses.course_name
FROM students
JOIN courses ON students.course_id = courses.course_id
WHERE students.name LIKE ? OR students.reg_no LIKE ?
ORDER BY students.id DESC";

$stmt = $conn->prepare($query);
$search_term = "%$search%";
$stmt->bind_param("ss", $search_term, $search_term);
$stmt->execute();
$result = $stmt->get_result();
?>
```

9.

UI Design (Bootstrap & Responsiveness)	Designs clean, responsive, and user-friendly interface using Bootstrap components effectively	Implements responsive design with acceptable layout	Basic Bootstrap used but inconsistent layout	Interface is not responsive; poor layout	2	
--	---	---	--	--	---	--

Bootstrap Link:

```
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
```

Login Page UI

Login

Username

Password

Login

Dashboard Page UI

Student MS

Manage Students Logout

Welcome, admin (Admin)

Total Students: 4

Total Courses: 9

Manage Students Page UI

Student Records

Search by name or Reg No...

Search

+ Add Student

Reg No	Name	Email	Course	Actions
20DIT24F1079	MOHAMAD ZULFITRY BIN RAMLI	zulfitry@gmail.com	Diploma of Information Technology	Edit Delete
20DIT24F1008	WAWAN BIN ABDULLAH	wawanabduallah1704@gmail.com	Diploma of Information Technology	Edit Delete
20DIT24F1012	AWANGKU NURFIKRI BIN AWANGKU SOPHIAN	awangkunurfikri@gmail.com	Diploma of Information Technology	Edit Delete
20DPM24F1078	SADIQ SEGARAGA	sadiqsegaraga@gmail.com	Diploma in Business Studies	Edit Delete

[Back to Dashboard](#)

10.

Error Handling & Validation	Integrates comprehensive validation and structured error handling across system	Implements proper validation and error messages	Basic validation with limited error handling	No validation or error handling implemented	2	
-----------------------------	---	---	--	---	---	--

<<edit_student.php>>

```
if (isset($_POST['update'])) {
    $reg = $_POST['reg_no'];
    $name = $_POST['name'];
    $email = $_POST['email'];
    $course = $_POST['course_id'];
    $student_id = $_POST['id'];

    $update_stmt = $conn->prepare("UPDATE students SET reg_no=?, name=?,
email=?, course_id=? WHERE id=?");
    $update_stmt->bind_param("sssi", $reg, $name, $email, $course,
$student_id);

    if ($update_stmt->execute()) {
        header("Location: manage_students.php?success=StudentUpdated");
        exit();
    } else {
        $error = "Error updating record: " . $conn->error;
    }
}
```

<<login.php>>

```
if ($result->num_rows == 1) {
    $user = $result->fetch_assoc();
    $_SESSION['username'] = $username;
    $_SESSION['role'] = $user['role'];
    header("Location: dashboard.php");
} else {
    $error = "Invalid Username or Password";
}
?>
```